

HOMEWORK 1

35 points

1. Which of the following are *mathematical statements*? $4 \times 1 = 4$ points

a) $2+3=5$ ✓

b) $x+2=7$ ✗

c) "Paris is the capital of France." ✓

d) $4 < y + 2$ ✗

a, c

2. Let P: "7 is a prime number" and Q: "2 is an even number." $1 + 1 = 2$ points

"and" - both True or it is False

- Write the conjunction $P \wedge Q$
- Is it true or false?

True

True

- 7 is a prime number and 2 is an even number.
- True

3. Let P: "5 is divisible by 2" and Q: "9 is greater than 7." $1 + 1 = 2$ points

disjunction - "or" - either has to be True

- Write $P \vee Q$
- Determine its truth value.

False

True

- 5 is divisible by 2 or 9 is greater than 7
- True

4. Write the following statement using mathematical notations:

“Some integers are negative.”

Negate the above statement. Use mathematical notations in your final answer.

2 points

$\exists x \in \mathbb{Z}. x < 0 \rightarrow \text{original}$

$\neg (\exists x \in \mathbb{Z}. x < 0)$

$\forall x \in \mathbb{Z}. x \geq 0$

5. Let P be False, Q : True and R : True
by 5." Find the truth value of the compound statement: $(\neg P \wedge Q) \vee R$. **3 points**

Show your steps and reasoning

$$(\neg P \wedge Q) \vee R$$

$$(\neg F \wedge T) \vee T$$

$$(T \wedge T) \vee T$$

$$T \vee T \rightarrow \text{True}$$

6. Express the following in words: **2 + 2 = 4 points**

- $\forall x \in \mathbb{N}. \exists y \in \mathbb{N}. (y > x)$
- $\exists x \in \mathbb{R}. \forall y \in \mathbb{R}. (x + y = 0)$

- For every natural number x , there exists a natural number y , such that $y > x$.
- There exists a real number x , such that for all real numbers y , $x + y = 0$.

7. Determine the truth value of the given statements: **3 * 1 = 3 points**

- $\forall x \in \mathbb{N}. \exists y \in \mathbb{N}. (x < y)$ **True**
- $\exists x \in \mathbb{N}. \forall y \in \mathbb{N}. (x < y)$ **False**
- $\exists x \in \mathbb{R}. (x^2 + 1 = 0)$ **False**

8. Construct Truth Tables for: $3 + 3 = 6$ points

a) $(P \wedge Q) \vee \neg R$

b) $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$

a)

P	Q	R	$\neg R$	$P \wedge Q$	$(P \wedge Q) \vee \neg R$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	T	F	T
F	T	T	F	F	F
F	T	F	T	F	T
F	F	T	F	F	F
F	F	F	T	F	T

b)

P	Q	R	$(P \Rightarrow Q)$	$(Q \Rightarrow R)$	$(P \Rightarrow Q) \wedge (Q \Rightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

True

P

$\Rightarrow$

Q

9. Consider the statement: “if a number is divisible by 4, it is even” $3 + 3 = 6$
points

- Write the converse, inverse and contrapositive of the above statement
- Which of the above are True and which are False?

Converse: $Q \Rightarrow P$ False

If a number is even, it is divisible by 4

Inverse: $\neg P \Rightarrow \neg Q$ False

If a number is not divisible by 4, it is not even

Contrapositive: $\neg Q \Rightarrow \neg P$ True

If a number is not even, it is not divisible by 4

10. Determine the truth value of each statement (True/False), assuming the universe is all integers: **3 points**

- If n is divisible by 6, then n is divisible by 3. **True**
- If n is divisible by 3, then n is divisible by 6. **False**
- A number is divisible by 6 if and only if it is divisible by both 2 and 3. **True**